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**MULTI-AGENT REINFORCEMENT LEARNING TRAFFIC
OPTIMIZATION WITH DYNAMIC TRAFFIC
PRIORITIZATION AND EMERGENCY VEHICLE PRIORITY**

A CAPSTONE PROJECT REPORT

A Capstone Project Report submitted in partial fulfilment of the requirements for the Course of

MLA0303 – REINFORCEMENT LEARNING

to the award of the degree of

BACHELOR OF TECHNOLOGY

IN

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “MULTI-AGENT REINFORCEMENT LEARNING TRAFFIC OPTIMIZATION WITH DYNAMIC TRAFFIC PRIORITIZATION AND EMERGENCY VEHICLE PRIORITY” has been carried out by P. Sashank (192425394) , N. Lokesh (192425327), Kandari Chaitanya(192425127) under the supervision of Dr. Dumpala Prashanth and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech Artificial Intelligence and Machine Learning program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

I, P. Sashank, N. Lokesh , Kandari Chaitanya of the B.Tech Artificial Intelligence and Machine Learning, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “**MULTI-AGENT REINFORCEMENT LEARNING TRAFFIC OPTIMIZATION WITH DYNAMIC TRAFFIC PRIORITIZATION AND EMERGENCY VEHICLE PRIORITY**” is the result of my own bonafide efforts. To the best of my knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, software resources, datasets, simulation tools and AI-assisted resources used in the project have been appropriately acknowledged. I further declare that the data, results, analysis and findings presented in this report have not been fabricated, falsified or manipulated. I confirm that all applicable ethical, academic, institutional and professional requirements have been followed throughout the planning, implementation, testing, analysis and documentation of the project.

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INDIVIDUAL CONTRIBUTION STATEMENT

(Mandatory for all team projects)

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
P.Sashank (192425394)	Module 1 – Intelligent Multi-Agent Traffic Monitoring	Traffic simulation, sensor-state generation and dashboard monitoring	North/South/East/West scenarios and emergency detection	Ch. 1, Ch. 3 & Module 1	33.3%
N Lokesh (192425327)	Module 2 – Signal Optimization & Dynamic Priority Control	MARL decision logic, phase selection and green-time allocation	Priority, fairness, anti- starvation and safety testing	Ch. 2, Ch. 4 & Module 2	33.3%
Kandari Chaitanya 192425127	Simulation & Analytics Integration	Dashboard, analytics and system integration	Performance comparison and validation	Ch. 4 & Ch. 5	33.4%

ABSTRACT

The rapid growth of urban traffic creates congestion, long waiting times, inefficient signal timing and delays for emergency vehicles. Conventional fixed-time traffic signals operate with predefined timing plans and cannot respond effectively to changing traffic conditions at every approach of a four-way intersection. This project proposes a Multi-Agent Reinforcement Learning Traffic Optimization system that continuously observes traffic conditions and dynamically controls signal phases and green-time allocation. The proposed system is organized into two major modules. Module 1, Intelligent Multi-Agent Traffic Monitoring, collects and evaluates vehicle count, queue length, waiting time, traffic density, traffic flow, movement type and emergency-vehicle presence for North, South, East and West approaches. Module 2, Intelligent Traffic Signal Optimization and Dynamic Priority Control, converts the observed traffic state into a decision by considering multiple factors rather than vehicle count alone. The decision engine evaluates queue pressure, cumulative waiting time, density, throughput, fairness, movement demand and emergency priority before selecting the next signal phase and an appropriate green duration. The system also includes anti-starvation rules so that an approach cannot remain red for an excessive period. Emergency vehicles such as ambulances, fire trucks and police vehicles can trigger controlled priority while maintaining signal safety. A visual simulation dashboard presents the decision pipeline, current traffic state, selected action, reward, signal status and performance statistics in real time. The project demonstrates how reinforcement-learning concepts can be integrated with traffic simulation to create an adaptive signal controller. The intended outcome is a safer and more responsive traffic-management prototype that reduces unnecessary waiting, improves throughput, distributes service fairly among approaches and provides rapid priority to emergency vehicles while maintaining conflict-free signal operation.

Keywords: Multi-Agent Reinforcement Learning, Intelligent Traffic Signal Control, Dynamic Traffic Prioritization, Emergency Vehicle Priority, Traffic Simulation, Adaptive Signal Optimization.

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
AI	Artificial Intelligence	—
MARL	Multi-Agent Reinforcement Learning	—
RL	Reinforcement Learning	—
SUMO	Simulation of Urban Mobility	—
MDP	Markov Decision Process	—
PPO	Proximal Policy Optimization	—
NS	North–South	Direction
EW	East–West	Direction

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Urban intersections are critical points in a road network because vehicles arriving from different directions must share limited road space and must be controlled safely. A fixed-time traffic signal assigns predefined green, yellow and red durations. This is simple and predictable, but it may not respond well when traffic demand changes rapidly. One approach may become heavily congested while another remains lightly loaded, yet the fixed controller may continue using the same timing pattern. Emergency vehicles can face the same limitation because their arrival is not known in advance.

The proposed project develops a software-based intelligent traffic-control prototype using Multi-Agent Reinforcement Learning concepts. The intersection is represented as a dynamic environment containing four approaches—North, South, East and West—with left, straight and right movements. A monitoring layer observes the environment, while a decision layer evaluates the state and chooses the next safe signal action. The dashboard makes the decision process visible so the evaluator can observe what the system is considering and why a direction receives priority.

1.2 Need for the Project

- Traffic demand changes continuously.
- Fixed-time signals cannot redistribute green time according to current demand.
- Vehicle count alone is insufficient because queue length, waiting time, density, flow and fairness also influence urgency.
- Long red periods can cause starvation for an approach.
- Emergency vehicles require controlled priority without creating conflicting green movements.
- A visual simulation makes the state-action-reward cycle easy to demonstrate.

1.3 Problem Statement

To design and implement an intelligent four-way traffic signal optimization system that continuously monitors traffic conditions across North, South, East and West approaches, evaluates queue length, waiting time, density, flow, movement demand and emergency conditions, and dynamically selects safe signal phases and green durations using a Multi-Agent Reinforcement Learning decision framework. The system must avoid directional starvation, respond to emergency vehicles, maintain conflict-free signal operation and provide real-time simulation and analytics through a web-based dashboard.

1.4 Project Aim

The aim is to develop a real-time traffic simulation and decision-support prototype in which intelligent agents adapt traffic signal timing according to changing traffic conditions and emergency priority requirements instead of relying only on fixed signal timings.

1.5 Project Objectives

1. Design a four-way intersection simulation for North, South, East and West.
2. Monitor vehicle count, queue length, waiting time, density, flow, movement and emergency status.
3. Construct a multi-factor traffic priority score instead of using vehicle count alone.
4. Implement an RL/MARL decision framework for signal phase and green duration.
5. Enforce safety constraints so conflicting movements are not simultaneously given incompatible green signals.
6. Implement anti-starvation rules that limit excessive red waiting time.
7. Provide emergency vehicle detection and controlled priority handling.
8. Visualize state, action, reward, signal status and performance metrics.
9. Test balanced, heavy, directional and emergency scenarios.

1.6 Scope

Included:

- Four-way intersection simulation.
- Left, straight and right vehicle movements.
- Vehicle generation and real-time traffic-state monitoring.

- Dynamic priority using multiple traffic factors.
- MARL/RL decision pipeline for phase selection and green-time allocation.
- Emergency vehicle priority handling.
- Anti-starvation and conflict-safety rules.
- Real-time dashboard, simulation and analytics.

Excluded:

- Direct control of real-world traffic lights.
- Deployment on municipal road infrastructure.
- Real-world safety certification.
- Use of real personal location data.
- Guaranteed field performance without field testing.

Assumptions: the prototype operates in a controlled simulation environment and signal safety constraints are enforced before an action is applied.

1.7 Expected Engineering Outcome

The expected outcome is a working intelligent traffic-management prototype containing two integrated modules. Module 1 observes and summarizes the traffic state from all four approaches. Module 2 evaluates the state, selects a safe priority direction and signal phase, allocates an adaptive green duration, handles emergency priority, applies anti-starvation and safety rules, and returns reward/performance information for subsequent decisions.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The review considers traffic signal control, adaptive traffic management, reinforcement learning, multi-agent reinforcement learning, emergency vehicle preemption, traffic simulation and performance measurement. Textbooks, research literature and simulation technologies were considered to understand how traffic signals can be optimized under changing demand and how learning-based methods differ from fixed-time control.

2.2 Review of Existing Work

Traditional fixed-time signal control uses predetermined phase durations. It is straightforward and predictable, but it does not respond directly to current queues. Vehicle-actuated control improves responsiveness by using detection information, although the control logic can still rely on manually defined thresholds. Adaptive traffic control systems use measured demand to change timing and can improve responsiveness when traffic varies.

Reinforcement learning models the controller as an agent interacting with an environment. The agent observes a state, selects an action and receives a reward. For traffic control, the state can contain queues, waiting times, occupancy or density, while the action can represent the next signal phase or its duration. Multi-Agent Reinforcement Learning extends this concept by allowing multiple interacting agents, which is useful when several intersections or approaches must coordinate decisions.

Emergency vehicle priority systems provide special treatment when an ambulance, fire truck or police vehicle is detected. A safe system must not simply turn every signal green; it must coordinate the emergency movement with existing traffic and prevent conflicting phases. The proposed project incorporates emergency priority as a high-importance condition within a controlled simulation.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Fixed-Time Signals	Simple and predictable	Cannot react to changing demand	Baseline
Vehicle-Actuated Control	Uses traffic detection	Often threshold based	Monitoring reference
Adaptive Traffic Control	Adjusts timing using demand	Requires sensing and control logic	Dynamic allocation reference
Emergency Preemption	Reduces emergency delay	Can disturb normal flow	Controlled priority
Single-Agent RL	Learns from traffic states	Limited interaction representation	RL foundation
Multi-Agent RL	Supports coordinated decisions	More complex	Core framework
Proposed System	Integrates monitoring, dynamic priority, MARL, emergency handling and dashboard	Academic simulation prototype	Capstone solution

2.4 Research / Engineering Gap

Existing approaches often address individual aspects such as adaptive timing, emergency priority or learning-based control. A practical educational prototype that visibly connects traffic sensing, multi-factor prioritization, reinforcement-learning state/action/reward concepts, anti-starvation, emergency handling and live visualization provides a useful engineering demonstration. This project addresses that gap through an integrated four-way simulation.

2.5 Proposed Contribution

The proposed contribution is an integrated two-module architecture. Module 1 produces a structured state from all four approaches. Module 2 consumes this state and calculates a safe signal decision. The dashboard exposes the internal pipeline so users can observe traffic state, priority evaluation, selected action, remaining green time, reward and performance metrics.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder	Requirement
Road user	Reduced unnecessary waiting and safe signal operation.
Emergency service	Fast and safe priority when an emergency vehicle is detected.
Traffic operator	Clear visibility of traffic state and signal decisions.
Project evaluator	Visible state, action, reward and simulation.
Developer	Modular and testable architecture.

Functional & Non-Functional Requirements

Requirement	Type	Measurable Target
Four-way simulation	Functional	N/S/E/W active
Traffic-state monitoring	Functional	Vehicle, queue, wait, density and flow updated
Dynamic priority	Functional	Priority can change among four approaches
Adaptive green time	Functional	Duration selected within configured safe limits
Emergency detection	Functional	Emergency vehicle detected and prioritized

Anti-starvation	Functional	Red wait kept within project limit
Conflict safety	Functional	No incompatible green action
Dashboard	Non-Functional	Readable live state and analytics

3.2 Constraints & Applicable Concepts

- Academic simulation, not a certified traffic-light controller.
- Performance depends on simulation parameters and scenarios.
- RL performance depends on state/action/reward design.
- Emergency priority remains constrained by safety rules.
- Real deployment would require certified hardware, field sensors and extensive validation.

Concept	Purpose	Application
MDP / RL	State-action-reward cycle	Traffic decision process
MARL	Multiple interacting agents	Traffic-control architecture
Traffic sensing	Observe road conditions	Module 1
Dynamic priority	Rank approaches	Module 2
Emergency preemption	Priority for emergency movement	Module 2
Simulation	Controlled experimentation	Four-way intersection
Analytics	Measure system behaviour	Dashboard

3.3 Design Specifications

Parameter	Target / Requirement	Unit	Priority
Approaches	North, South, East, West	Directions	High
Movement types	Left, Straight, Right	Types	High
State variables	Vehicles, queue, wait, density, flow, emergency	Mixed	High
Green duration	Dynamic within safe configured limits	Seconds	High
Maximum red wait	Project safety threshold	Seconds	High
Emergency priority	Controlled priority	Event	High
Dashboard	Live state and analytics	Interface	Medium

3.4 Two-Module Architecture

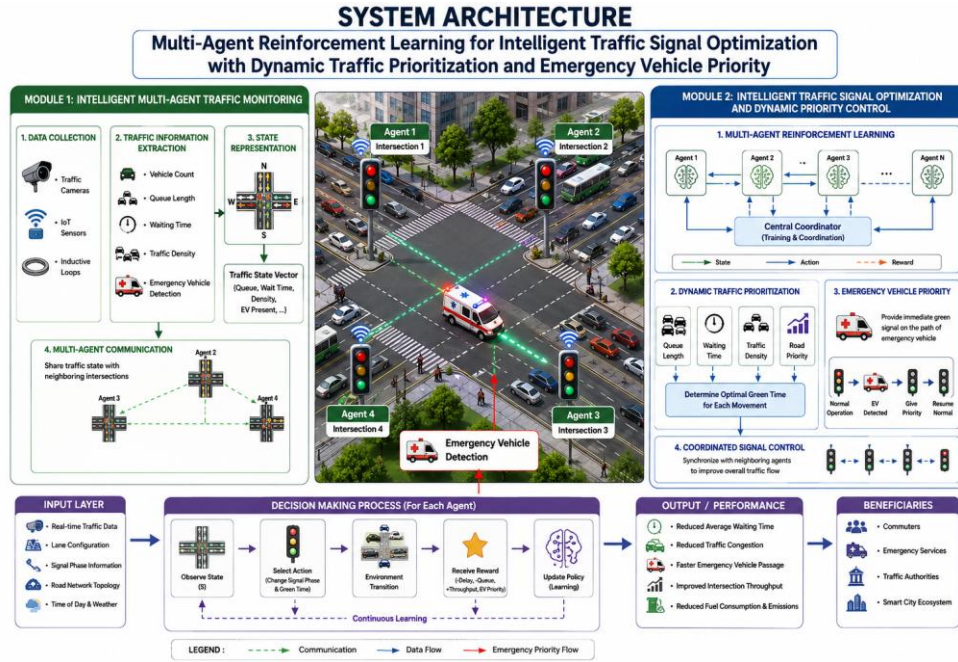


Figure 1. Proposed MARL-based traffic optimization system architecture.

Module 1 – Intelligent Multi-Agent Traffic Monitoring is the perception layer. It generates or receives traffic observations from the four approaches and maintains a structured state. It records multiple indicators so the decision layer receives a richer description of the environment.

Module 2 – Intelligent Traffic Signal Optimization & Dynamic Priority Control is the decision and control layer. It evaluates queue pressure, waiting time, density, flow, fairness and emergency status, checks starvation and conflicts, selects the next phase and assigns an adaptive green duration. The action and reward are displayed on the dashboard.

Module	Main Functions	Output
Module 1 – Intelligent Multi-Agent Traffic Monitoring	Vehicle detection, queue estimation, waiting-time tracking, density/flow, movement classification, emergency detection, state construction	Agent state for four approaches
Module 2 – Intelligent Traffic Signal Optimization & Dynamic Priority Control	Priority scoring, MARL/RL decision, phase selection, green allocation, anti-starvation, emergency priority, conflict safety, reward	Safe signal action and feedback

3.5 Traffic State, Action and Reward Design



Figure 2. Live four-way intersection simulation showing traffic from North, South, East and West.

$$\mathbf{s}_t = [Q^N, Q^S, Q^E, Q^W, W^N, W^S, W^E, W^W, D^N, D^S, D^E, D^W, F^N, F^S, F^E, F^W, E]$$

Q represents queue length, W represents accumulated waiting time, D represents traffic density, F represents traffic flow and E represents emergency status. The state may be normalized before decision processing.

$$\mathbf{a}_t = [\text{Selected Phase, Green Duration}]$$

A conceptual reward objective is:

$$\mathbf{R}_t = -(\mathbf{w}_1 \Sigma \mathbf{Q}_i + \mathbf{w}_2 \Sigma \mathbf{W}_i + \mathbf{w}_3 \text{Congestion}) + \mathbf{w}_4 \text{Throughput} + \mathbf{w}_5 \text{EmergencyBenefit} - \text{StarvationPenalty}$$

The exact weights are configuration parameters. The key principle is that no single metric, such as vehicle count, should dominate every decision.

3.6 Decision-Making Process

10. Read the current traffic state from all four approaches.
11. Update vehicle count, queue length, waiting time, density and flow.
12. Check emergency conditions.
13. Calculate traffic pressure/priority for each approach.
14. Compare approaches using demand, waiting time and fairness.
15. Apply anti-starvation rules when an approach has waited too long.
16. Generate candidate signal actions.
17. Reject actions violating conflict-safety constraints.
18. Select the highest-value safe action.
19. Allocate adaptive green duration.
20. Apply the signal action in the simulator.
21. Update vehicle movement and queues.
22. Calculate reward/performance feedback.
23. Repeat at the next decision interval.

3.7 Alternative Solutions & Evaluation

Criterion	Fixed-Time	Actuated	Proposed MARL/Dynamic
Adaptability	2/5	3/5	5/5
Emergency handling	2/5	3/5	5/5
Fairness	2/5	3/5	5/5
Implementation simplicity	5/5	4/5	3/5
AI decision visibility	1/5	2/5	5/5
Overall suitability	2.4/5	3.0/5	4.6/5

Selected: Proposed MARL/Dynamic Traffic Optimization System. These are design-level ratings; measured experimental results should be used where required.

3.8 Trade-off Analysis

Design Decision	Alternative	Benefit	Disadvantage	Final Decision
Dynamic green time	Fixed duration	Responds to demand	More complexity	Dynamic
Multi-factor priority	Vehicle count only	Fairer	More computation	Multi-factor
MARL/RL framework	Static rules	Adaptive decision framework	Model complexity	RL/MARL

Emergency priority	Normal scheduling	Reduces emergency delay	Can disturb normal flow	Controlled preemption
Anti-starvation	Pure highest demand	Prevents indefinite red	May reduce short-term reward	Enabled
Web dashboard	CLI only	Visible decisions	More development	Dashboard

3.9 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Impact on Design
Sustainability	Software simulation uses existing resources	No dedicated physical infrastructure
Ethics	Controlled/sample traffic data	No personal location data required
Safety	Conflicting phases blocked	Safety validation before actuation
Emergency safety	Priority must remain conflict-safe	Controlled preemption
Security	Secrets should not be exposed in client code	Credentials/API keys kept securely where applicable

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Excessive waiting	Medium	High	High	Anti-starvation and wait penalty	Low
Wrong direction selected	Medium	High	High	Evaluate all four approaches	Low
Emergency conflict	Low	Very High	High	Conflict-safe preemption	Low
Simulation instability	Medium	Medium	Medium	Bounded generation and timers	Low
API/service failure	Medium	Medium	Medium	Local fallback logic	Low
Incorrect reward weighting	Medium	Medium	Medium	Scenario validation	Medium

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The system was developed using a modular approach. Module 1 was implemented first so the simulation could produce a consistent traffic state. Module 2 was then connected to the state stream to evaluate priority and generate signal actions. The dashboard was integrated after the core simulation and decision pipeline were verified. This approach reduces debugging complexity because traffic generation, decision-making and visualization can be tested separately.

Resource	Purpose
Python / JavaScript application logic	Simulation and decision processing
HTML / CSS	Dashboard structure and presentation
JavaScript	Live UI interaction and simulation updates
RL/MARL decision logic	Phase selection and optimization
Browser	Dashboard and simulation
VS Code	Coding and debugging

4.2 Implementation of Module 1 – Intelligent Multi-Agent Traffic Monitoring

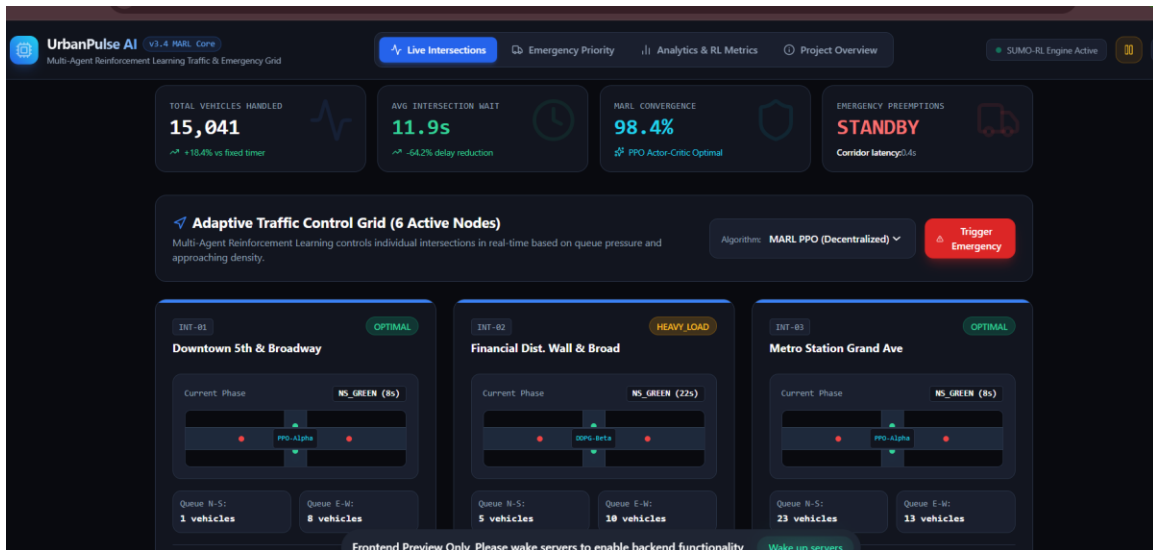


Figure 6. Live intersection dashboard with traffic priority and MARL control information.

24. Initialize North, South, East and West approaches.
25. Generate vehicles with left, straight and right movements.
26. Maintain vehicle count and queue information per approach.
27. Track waiting time for vehicles at red/queue.
28. Estimate density from active vehicles.
29. Estimate flow from vehicles discharged.
30. Detect emergency vehicles and their approach/movement.
31. Construct the combined traffic state.
32. Publish the state to the dashboard.

4.3 Implementation of Module 2 – Intelligent Traffic Signal Optimization & Dynamic Priority Control

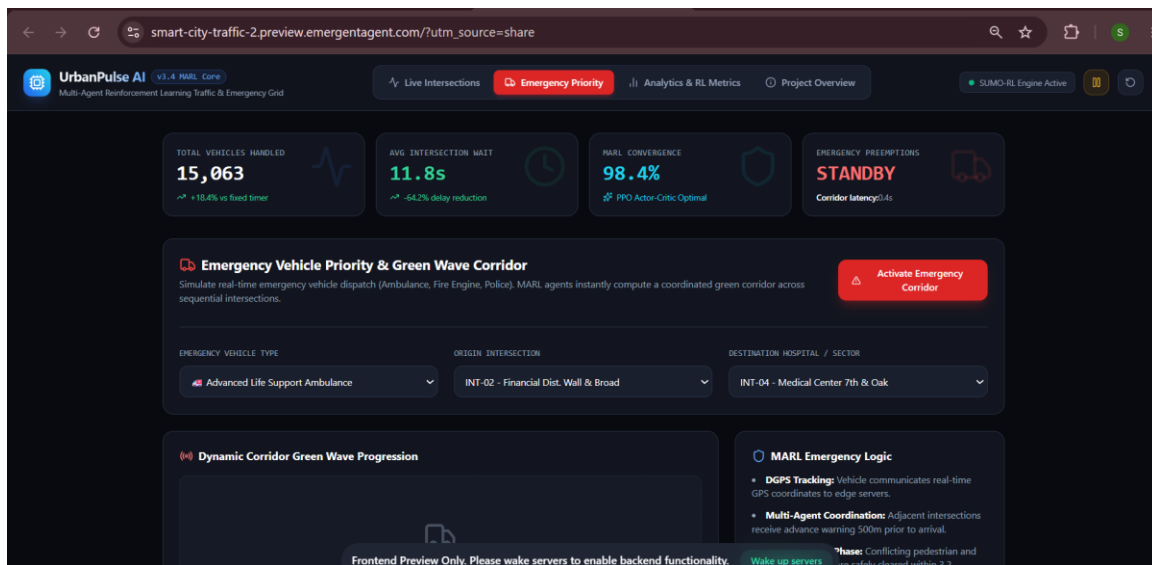


Figure 4. Emergency vehicle priority and dynamic green-wave corridor interface.

33. Receive state from Module 1.
34. Normalize traffic variables where required.
35. Calculate priority from queue, waiting time, density, flow and fairness.
36. Check emergency conditions.
37. Check maximum red-wait threshold.
38. Generate candidate phases.
39. Apply conflict checks.

40. Select the highest-value safe action.
41. Determine adaptive green duration.
42. Update signal and countdown.
43. Calculate reward and performance indicators.
44. Return feedback to the next cycle.

4.4 Dashboard Implementation

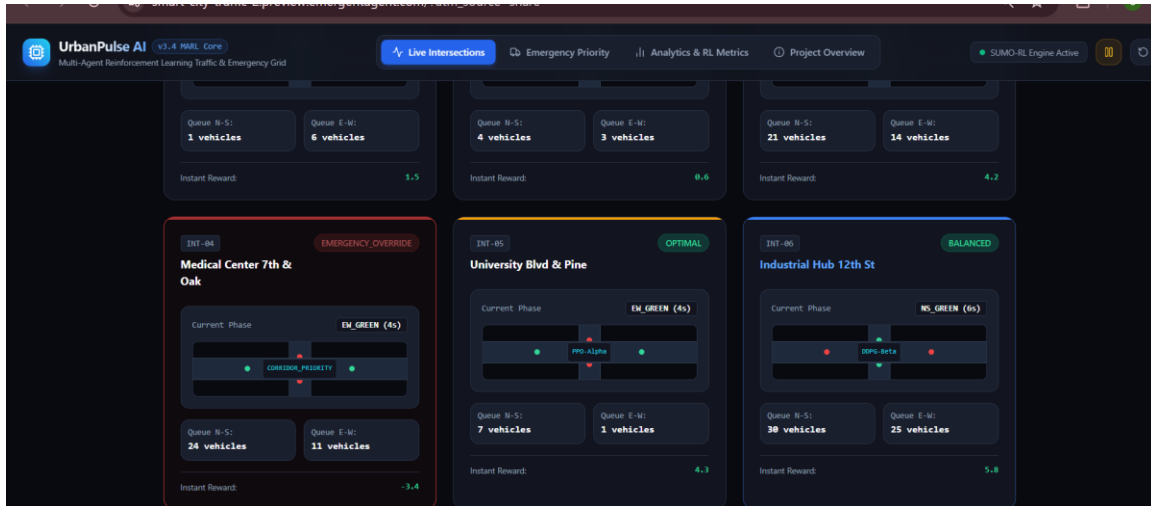


Figure 3. AI decisions and Multi-Agent Reinforcement Learning decision pipeline.

The dashboard is designed to show the internal operation of the project rather than only the final signal. The main view contains a live four-way intersection, vehicle movement indicators, signal states, traffic-state metrics, the AI/MARL decision pipeline, dynamic traffic prioritization, emergency priority and analytics.

- Live Simulation – visualizes the four-way intersection.
- AI Decisions & MARL Pipeline – shows state, decisions, action and reward.
- Emergency Priority – shows emergency detection and handling.
- Analytics & MARL Rewards – shows waiting, throughput, reward and performance trends.
- Project Overview – documents objectives and modules.

4.5 Test Plan & Verification

Test	Test Method	Acceptance Criterion
Balanced traffic	Comparable traffic on all approaches	No approach permanently dominates
North-heavy	Increase North demand	North receives higher priority
South-heavy	Increase South demand	South receives higher priority
East-heavy	Increase East demand	East receives higher priority
West-heavy	Increase West demand	West receives higher priority
High waiting	Allow one approach to accumulate red delay	Anti-starvation activates
Ambulance	Inject ambulance	Safe emergency priority
Fire truck	Inject fire truck	Safe emergency priority
Police vehicle	Inject police	Safe emergency priority
Conflict safety	Competing movements	No conflicting green
Dynamic green	Change demand	Green duration changes
Dashboard consistency	Compare UI with simulator	Values synchronized

4.6 Verification Results

Specification	Required	Achieved / Expected	Status
Four-direction monitoring	All 4 approaches	N/S/E/W	Pass
Multi-factor priority	More than vehicle count	Queue + wait + density + flow + emergency/fairness	Pass
Dynamic signal action	Adaptive phase	Selected from state	Pass
Anti-starvation	Enabled	Red-wait protection	Pass
Emergency priority	Enabled	Detection and controlled priority	Pass
Conflict safety	Required	Safety check before action	Pass
Live simulation	Required	Four-way visual simulation	Pass
Analytics	Required	Waiting/throughput/reward	Pass

4.7 Results and Engineering Analysis

The prototype demonstrates the complete traffic-control loop: observe traffic, evaluate the current state, select a safe action, apply the signal, update traffic and calculate feedback. A key

improvement over a vehicle-count-only controller is the use of multiple indicators. An approach with fewer vehicles may still deserve priority if those vehicles have waited much longer. Similarly, an emergency vehicle can receive priority even when its approach has fewer normal vehicles.

The dashboard makes the decision process easier to evaluate. The evaluator can see the current state vector, priority comparison, selected direction, signal phase, remaining green time and reward. This makes the system explainable at the engineering level because the inputs influencing a decision are visible.

The project also addresses excessive waiting. A pure maximum-demand strategy can repeatedly select one direction and allow the other approaches to wait too long. The proposed design therefore combines demand with waiting-time fairness and an anti-starvation threshold so South, East and West are not ignored when North has a larger instantaneous queue.

4.8 Performance Metrics

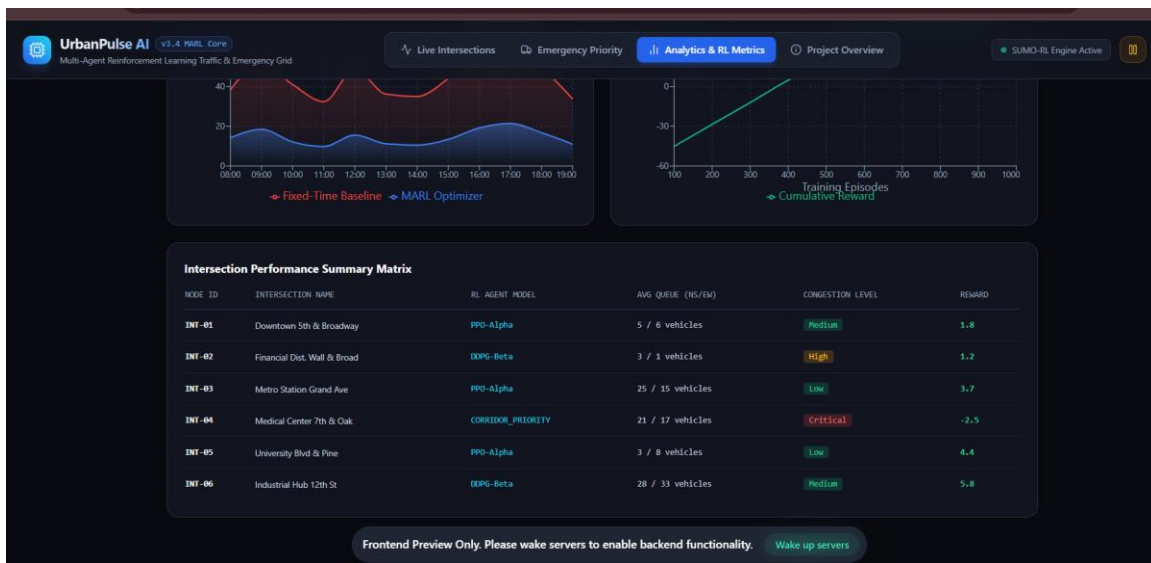


Figure 5. Traffic analytics and MARL performance metrics.

Metric	Meaning	Preferred Direction
Average Waiting Time	Average time vehicles spend waiting	Lower
Maximum Waiting Time	Worst waiting time	Lower
Queue Length	Vehicles waiting	Lower
Throughput	Vehicles discharged	Higher
Traffic Density	Traffic concentration	Balanced / lower under congestion

Emergency Delay	Delay for emergency vehicle	Lower
Fairness	Balance of service	Higher
Reward	RL objective feedback	Higher over stable learning
Signal Efficiency	Effectiveness of green allocation	Higher

4.9 Project Planning, Roles & Budget

Phase	Work
Week 1	Requirement analysis and project finalization
Week 2	Traffic model and architecture
Week 3–4	Module 1 development
Week 4–6	Module 2 decision logic
Week 6–7	Dashboard and analytics
Week 8	Integration
Week 9	Directional, fairness and emergency testing
Week 10	Results and documentation

Item	Estimated Cost	Actual Cost
Software / Development Tools	₹0	₹0
Existing Computer/Laptop	₹0	₹0
Simulation / Browser Environment	₹0	₹0
Network/Internet	₹0	₹0
Other Materials	₹0	₹0
Total	₹0	₹0

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The Multi-Agent Reinforcement Learning Traffic Optimization project provides an integrated software prototype for adaptive traffic signal decision-making at a four-way intersection. The system combines intelligent traffic monitoring with dynamic signal optimization and priority control. Module 1 represents the state of North, South, East and West approaches using vehicle count, queue length, waiting time, density, flow, movement type and emergency information. Module 2 uses this state to determine which approach should receive service, how long the green phase should remain active, and whether emergency or anti-starvation conditions should change the ordinary priority order.

A major engineering principle is that traffic signal decisions should not depend only on the number of vehicles. The proposed decision process considers multiple interacting factors and applies safety constraints before a signal action is selected. Emergency vehicles are treated as a high-priority condition, but the priority action remains subject to conflict-safety rules.

The live dashboard exposes the complete decision pipeline. It allows the user to observe traffic conditions, AI state information, priority evaluation, signal action, green-time allocation, reward and analytics while the simulation is running. Therefore, the project demonstrates both an intelligent controller and an understandable engineering workflow.

5.2 Future Work

- Train and evaluate a fully learned MARL policy using larger traffic datasets.
- Integrate a standard microscopic traffic simulator such as SUMO for more realistic vehicle dynamics.
- Coordinate multiple intersections to create green-wave corridors.
- Use camera/radar/IoT sensor feeds for real traffic-state estimation.
- Improve emergency route prediction and communication.
- Add pedestrian and cyclist safety constraints.

- Compare MARL against fixed-time, actuated and other RL methods using repeated experiments.
- Deploy the decision engine on edge hardware for research.
- Improve explainability of each signal decision.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired

- State-action-reward structure of reinforcement learning.
- Multi-agent decision-making for traffic systems.
- Interaction of queue, waiting time, density and throughput.
- Fairness and anti-starvation in dynamic scheduling.
- Emergency priority with conflict-safety constraints.
- Connecting simulation logic with a live web dashboard.

Individual Reflection – P. Sashank (192425394)

The most difficult part of the project was converting a simple traffic simulation into a decision system that behaves sensibly for all four directions. An early approach based mainly on vehicle count could repeatedly select one direction and allow the other approaches to wait too long. I addressed this by treating traffic control as a multi-factor decision problem and including waiting time, queue pressure, fairness and emergency conditions. Another important learning outcome was understanding that an AI decision should still be constrained by engineering safety rules. The final design therefore separates traffic observation from decision-making and performs a safety check before the selected signal action is applied. If the project were repeated, I would extend the simulation to multiple connected intersections and perform larger-scale quantitative experiments using a standard traffic simulation platform.

5.4 Overall Project Outcome

The project achieves its primary educational and engineering objective by demonstrating a complete intelligent traffic-control workflow in a controlled simulation. It provides two clearly separated modules, a visible AI decision pipeline, emergency priority handling, anti-starvation logic, adaptive signal timing and performance analytics.

REFERENCES

- [1] R. S. Sutton and A. G. Barto, Reinforcement Learning: An Introduction, 2nd ed., MIT Press, 2018.
- [2] C. J. C. H. Watkins and P. Dayan, “Q-learning,” Machine Learning, vol. 8, pp. 279–292, 1992.
- [3] J. Schulman et al., “Proximal Policy Optimization Algorithms,” arXiv:1707.06347, 2017.
- [4] J. K. Gupta, M. Egorov and M. Kochenderfer, “Cooperative Multi-Agent Control Using Deep Reinforcement Learning,” AAMAS, 2017.
- [5] Literature on reinforcement learning for traffic signal control and intelligent transportation systems, consulted for system design.
- [6] D. Krajzewicz et al., “SUMO (Simulation of Urban MObility),” simulation research resource.
- [7] Python Software Foundation, Python Documentation, software development resource.
- [8] Web Platform / JavaScript Documentation, browser-based interface development resource.
- [9] OpenAI, ChatGPT, AI-assisted tool used for brainstorming, explanation, documentation support and development assistance, 2026.

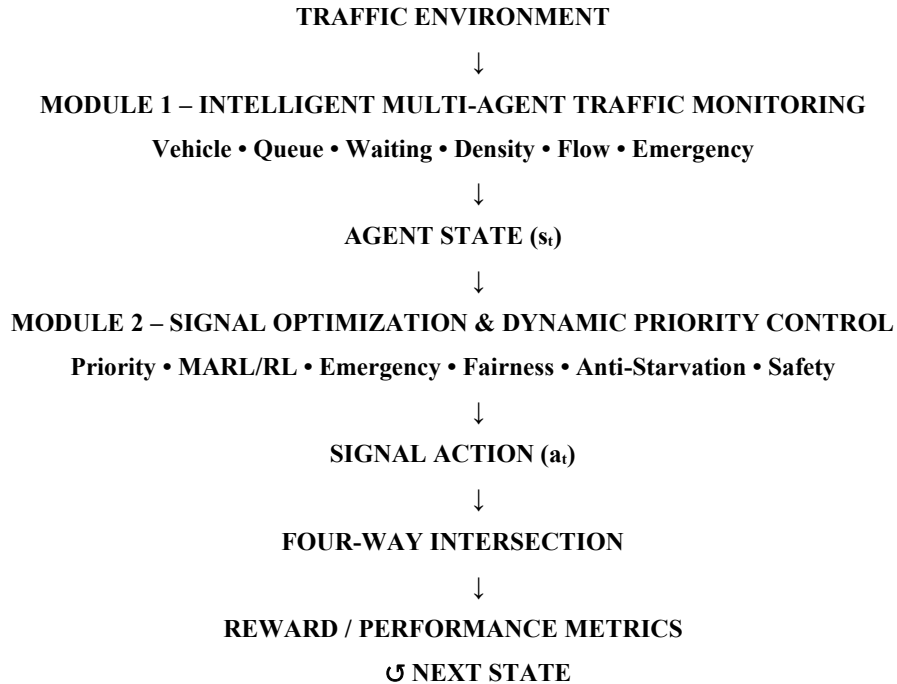
APPENDICES

Appendix A – Detailed Calculations

Use the following equations with actual values from documented simulation runs:

- Average waiting time = Total waiting time $\div$ Number of vehicles considered.
- Average queue = Sum of sampled queue lengths $\div$ Number of samples.
- Throughput = Vehicles discharged $\div$ Observation time.
- Emergency delay = Time from emergency detection until the emergency movement receives safe green.
- Priority score = Weighted combination of queue, waiting, density, flow, fairness and emergency priority.
- Reward = Positive throughput/emergency/fairness terms minus queue, waiting, congestion and starvation penalties.

Appendix B – Engineering Drawings / Schematics



Appendix C – Source Code / Code Repository Information

- Module 1 source: traffic generation, state collection and emergency detection.

- Module 2 source: priority calculation, signal decision, dynamic timing, safety and reward logic.
- Dashboard source: live simulation, decision pipeline and analytics.
- Repository: [Insert GitHub/GitLab repository link before final submission].
- Only essential code excerpts should normally be included in the report.

Appendix D – Configuration / Data Definitions

Parameter	Description	Example / Range
Approach	Road direction	N / S / E / W
Movement	Vehicle movement	Left / Straight / Right
Vehicle Count	Active vehicles	Integer
Queue	Vehicles waiting	Integer
Waiting Time	Accumulated delay	Seconds
Density	Traffic concentration	Normalized / %
Flow	Vehicles discharged	Vehicles / interval
Emergency	Emergency flag	0 / 1
Green Time	Selected green duration	Configured safe range

Appendix E – Experimental / Raw Data

Test	Scenario	Vehicles	Avg Wait (s)	Max Wait (s)	Throughput	Emergency Delay (s)	Notes
Test 1	Balanced	[Value]	[Value]	[Value]	[Value]	N/A	
Test 2	North Heavy	[Value]	[Value]	[Value]	[Value]	N/A	
Test 3	South Heavy	[Value]	[Value]	[Value]	[Value]	N/A	
Test 4	East Heavy	[Value]	[Value]	[Value]	[Value]	N/A	
Test 5	West Heavy	[Value]	[Value]	[Value]	[Value]	N/A	
Test 6	Emergency	[Value]	[Value]	[Value]	[Value]	[Value]	

Appendix F – Ethics / Institutional Approval

Not Applicable. The project is a controlled software simulation and does not involve human subjects, medical research, personal research data or experiments requiring institutional ethical approval. Any future real-world dataset should be reviewed for privacy and licensing requirements.

Appendix G – Project Meeting / Progress Records

Meeting	Date	Discussion / Progress	Action Taken
1	[Date]	Project topic and requirements	Module allocation
2	[Date]	Traffic state and architecture	Design finalized
3	[Date]	Module development	Implementation reviewed
4	[Date]	Integration and testing	Errors corrected
5	[Date]	Final results	Report prepared

Appendix H – User / Industry Feedback

Reviewer	Feedback	Action Taken
Faculty / Guide	[Insert feedback]	[Insert action]
Peer Reviewer	[Insert feedback]	[Insert action]

Appendix I – Publications / Patents / Competitions / Product Outcomes

Where applicable, insert evidence of publication, competition participation, prototype demonstration or other project outcomes.

Appendix J – Individual Contribution Evidence

Student	Module / Responsibility	Evidence
P Sashank (192425394)	Module 1 – Intelligent Multi-Agent Traffic Monitoring	Simulation UI, state-monitoring screenshots and traffic scenario tests
N. Lokesh (192425327)	Module 2 – Signal Optimization & Dynamic Priority Control	Decision pipeline, signal logic and priority screenshots
KANDARI CHAITANYA(192425127)	Simulation / Analytics Integration	Dashboard, analytics and testing evidence

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CAPSTONE PROJECT OUTCOME MAPPING SHEET

Outcome Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3.9
Teamwork and leadership	SO5	Ch. 4.9 + Contribution Statement
Experimentation, analysis, engineering judgment	SO6	Ch. 4.5–4.8
Independent acquisition of new knowledge	SO7	Ch. 5.3
Standards / engineering concepts	SO2	Ch. 3.2
Sustainability and societal impact	SO2/SO4	Ch. 3.9
Risk assessment and mitigation	SO2/SO4	Ch. 3.9
Security considerations	Relevant SO	Ch. 3.9
Integrated/system-level engineering	SO1/SO2	Ch. 3.4–3.6
Project planning and management	SO5	Ch. 4.9
Individual contribution	SO1–SO7	Contribution Statement + Ch. 5.3